



An EEG signal encryption algorithm based on dual-composite IT-ICMIC chaotic map and adaptive non-uniform partition

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ARTICLE INFO

Keywords:

EEG
Dual composite IT-ICMIC
Adaptive non-uniform partition
Chaotic map

ABSTRACT

Electroencephalogram (EEG) signals have become critical in applications such as medical diagnostics and neurofeedback systems. However, its sensitivity makes it vulnerable to unauthorized access and the risk of data leakage. To address these challenges, this study proposes an EEG signal encryption method based on a dual composite Inverse Trigonometric Iterative Chaotic Map (IT-ICMIC) and adaptive non-uniform partition. To enhance the security of EEG signal encryption, the dual composite IT-ICMIC chaotic map is introduced, addressing the limitations of traditional single chaotic maps in complexity and unpredictability. Additionally, the adaptive non-uniform partition algorithm explores the intrinsic dynamic characteristics of EEG signals. Further, the mined features integrate with the fundamental properties of EEG signals to generate chaotic sequences, enabling efficient and robust encryption. Extensive experiments and security analysis demonstrate that the proposed method achieves superior performance for EEG signal encryption, with an average NSCR of 100, a UACI of 33.36 highlighting its strong encryption effectiveness.

1. Introduction

Information and communication technologies (ICTs) provide diverse dissemination and opportunities to exchange health information. With the global outbreak of COVID-19, a large amount of sensitive healthcare data needs to be transmitted and stored over public networks to meet the needs of online healthcare and remote services [1]. However, unprotected communication exchanges not only exacerbate the security risks of healthcare data [2] but also make information security of healthcare data a critical issue that needs urgent attention. To this end, many teams have conducted extensive research toward secure data dissemination. Al-Hawawreh M. et al. [3] proposed a privacy-preserving framework combining data fusion and quantum deep learning techniques aimed at detecting cyber-attacks in IoT healthcare systems; Pankaj et al. [4] carried out a comprehensive analysis and comparison of chaos-based medical image encryption techniques. JA Michel-Macarty et al. [5] proposed a chaos-based multi-user telemedicine communication scheme to improve the security and integrity of the wireless transmission of electrocardiogram (ECG) signals in telemedicine.

As research into brain science continues to grow, the need to protect medical data extends further into neuroscience. This field not only puts higher requirements on the secure transmission and storage of data

but also places special emphasis on ensuring the privacy protection and compliant utilization of data in complex research environments. In this context, physiological signals, represented by EEG signals, urgently require the construction of an efficient and rigorous protection mechanism to ensure data integrity and security, given that they face the risk of illegal interception of personal privacy data [6,7].

Security protection for physiological data such as EEG signals is mainly realized by watermarking [8] and encryption [9]. Among them, encryption is widely used due to its directness and efficiency. Specifically, the encryption method hides the EEG signal features by transforming the collected EEG signal into complex and disordered noise through the permutation and diffusion mechanism. Robinson et al. [10] proposed a feature extraction method with the Advanced Encryption Standard (AES) algorithm to encrypt the EEG signals. Not only that, Yogalakshmi et al. [11] proposed a secure encrypted transmission scheme for brainwave data based on the AES algorithm, focusing on security protection at the signal transmission level. Although the AES algorithm can secure EEG signals to a certain extent, it still faces some limitations in practical applications. Since EEG signals contain a large amount of physiological information [12], it is vital to protect them comprehensively and precisely during encryption. However, the AES method may encounter computational efficiency or data integrity

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<https://doi.org/10.1016/j.bspc.2025.108879>

Received 3 March 2025; Received in revised form 14 September 2025; Accepted 5 October 2025

Available online 10 October 2025

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challenges. Therefore, exploring and developing encryption schemes that are more suitable for the characteristics of EEG signals has become an important issue that needs to be solved.

Among signal encryption techniques, chaos theory has been particularly widely used to secure communication of major events [13] and encryption of multiple types of images [14–16]. Its distinguishing and unique feature is its long-term unpredictability [17]. These methods rely on two key properties of chaos: sensitivity to initial conditions and long-term unpredictability. The chaotic maps generate high-entropy series and drive permutation–diffusion steps that cover the signal structure while keeping computation lightweight. This encryption method is well-suited for EEG data because EEG data contains rich physiological and cognitive information that poses privacy risks. At the same time, EEG is highly non-stationary and variable across individuals, which requires encryption schemes that can adapt to temporal dynamics and ensure robustness against various attacks.

Although EEG signal processing methods based on chaotic encryption have demonstrated certain advantages in EEG signal encryption, several problems still need to be solved [18]. First, most existing approaches rely on a single low-dimensional chaotic mapping, which lacks sufficient complexity and exhibits low sensitivity to initial conditions. These constraints increase vulnerability to brute-force and statistical attacks, especially in high-security biomedical applications. Second, EEG signals exhibit rich dynamic characteristics such as non-stationarity, inter-subject variability, and high entropy. However, current methods often treat EEG as a general time series, failing to utilize these unique properties to generate personalized and adaptive chaotic sequences. Third, existing schemes typically employ fixed encryption frameworks that do not adapt to the intrinsic structural differences of EEG signals across individuals or channels. This results in limited robustness and adaptability in practical clinical or wearable applications. These limitations collectively reduce the adaptability, security, and practical deployment potential of existing chaos-based EEG encryption algorithms.

To address these issues, an EEG signal encryption method based on dual composite IT-ICMIC chaotic mapping and adaptive non-uniform partition is proposed. The dual composite IT-ICMIC mapping is designed to enhance complexity and unpredictability. In this way, it strengthens security and improves resistance against attacks compared with traditional single chaotic mappings. Furthermore, the adaptive non-uniform partition algorithm is introduced to exploit the intrinsic dynamics of EEG signals. This allows the generation of personalized chaotic sequences that incorporate subject-specific and channel-specific features. By combining high chaotic complexity with EEG-driven adaptability, the proposed method significantly enhances the robustness and practicality of EEG encryption, offering a more secure solution for diverse application scenarios, including clinical practice and wearable devices.

The main contributions are:

- A dual composite IT-ICMIC chaotic map is proposed to overcome the low complexity and weak sensitivity of traditional chaotic maps. The generated chaotic sequences show higher complexity and stronger resistance to attacks.
- An adaptive non-uniform partition algorithm is designed to match the non-stationary and diverse characteristics of EEG signals. This improves adaptability and robustness in the encryption process.
- An EEG encryption framework is developed by combining the dual composite chaotic map with the adaptive partition algorithm. Experiments confirm better efficiency and security compared with existing methods.

The subsequent arrangement is as follows: Section 2 reviews the related work on EEG encryption methods based on the chaos theory. Section 3 introduces the dual composite IT-ICMIC chaotic mapping;

Part 4 presents the overall EEG signal encryption and decryption process; Part 5 conducts experimental validation for this method; and Parts 6 and 7 discuss and summarize the method, respectively.

For clarity, the main abbreviations used in this manuscript are summarized in Table 1.

2. Related work

2.1. Encryption methods based on the chaos theory

With the rapid digitalization and informatization of healthcare, intelligent technologies are increasingly integrated into clinical workflows and decision support systems [21,22]. In recent years, machine learning and intelligent systems have been widely applied in healthcare, including heart disease prediction [23–25] and arrhythmia classification [26]. These advances demonstrate the growing reliance on biomedical data, which in turn raises concerns about the secure storage and transmission of this information. Parallel progress in signal processing has enabled automated analysis of complex physiological signals to support tasks like epilepsy detection, sleep stage classification, and emotion recognition [27–31]. Beyond these, decomposition and wavelet methods have also been applied to different types of signals, demonstrating broad adaptability while reinforcing the clinical value of biomedical signal [32–34]. While these applications highlight the clinical and technological value of biomedical data, they also expose sensitive personal information, making privacy protection a critical issue. Here, chaos-based encryption has emerged as a promising approach because of its unpredictability and strong sensitivity to initial conditions.

In recent years, several encryption techniques based on chaotic systems have been developed by existing teams [43,44]. Table 2 summarizes representative chaos-based encryption methods. As shown in Table 2, chaos-based encryption methods have been applied across diverse domains, including images, audio, genetic data, and bio-signals. These studies demonstrate the generality of chaotic maps in generating secure and efficient encryption methods, with strengths such as randomness, unpredictability, and lightweight computation. However, existing methods still have clear limitations. Low-dimensional chaotic maps suffer from restricted key space, while many approaches lack adaptability to physiological signals such as EEG. Specifically, Lin et al.'s method is visually effective but constrained by a low-dimensional map, and Liu et al.'s hyperchaotic scheme adds complexity without addressing EEG-specific temporal dynamics or individual variability. These gaps highlight the need for EEG-oriented chaotic encryption methods that are both efficient and adaptive.

While these studies have advanced the application of chaotic systems to EEG encryption, they still suffer from notable limitations. Lin et al.'s method, though visually effective, employs a low-dimensional map with constrained key space. Liu et al.'s scheme introduces higher complexity via hyperchaos, but does not exploit EEG-specific temporal dynamics or support adaptive encryption tailored to individual variability.

3. Dual composite IT-ICMIC chaotic map

In this section, the definition of the proposed dual composite IT-ICMIC chaotic map is provided, and its chaotic dynamic characteristics are tested.

3.1. ICMIC

One-dimensional iterative chaotic map (ICMIC) [45] is proposed in Eq. (1).

$$x_{i+1} = \sin\left(\frac{r_1}{x_i}\right) \quad (1)$$

where r_1 is the control parameter which is in the range of $(0, +\infty)$.

Table 1
List of used abbreviations.

Abbreviation	Full form	Abbreviation	Full form
AES	Advanced Encryption Standard	BCMI	Brain-like Computing & Machine Intelligence
CHB	Children's Hospital Boston	COVID	Coronavirus Disease
DCLE	Double Chaotic Logistic Encryption Algorithm [19]	DNA	Deoxyribonucleic Acid
ECG	Electrocardiogram	EEG	Electroencephalogram
EMD	Empirical Mode Decomposition	ICMIC	Iterative Chaotic Map with Infinite Collapsing
IT	Inverse Trigonometric	JA	Joint Approximation
LE	Lyapunov Exponent	MIT	Massachusetts Institute of Technology
NSCR	Number of Sample Change Rate	MSE	Mean Squared Error
NUP	Non-Uniform Partition	OR	Exclusive OR
PCC	Pearson Correlation Coefficient	PRD	Percentage Residual Deviation
SEED	SJTU Emotion EEG Dataset	SNR	Signal-to-Noise Ratio
SP	Signal Processing	K-STBCCS	K-Sine-Transform-based Coupling Chaotic System [20]
UACI	Unified Average Changing Intensity	XOR	Exclusive OR Operation

Table 2
Summary of chaos-based encryption methods.

Author	Data type	Approach	Strengths	Limitations
Li et al. [35]	Image	Tent map	Simple, lightweight	Low-dimensional map, limited key space
Kumar et al. [36]	Audio	Dual chaotic maps + DNA coding	High robustness, secure transmission	Domain-specific, higher complexity
Li et al. [37]	Image	Chaos with plaintext correlation	Efficient	Limited validation on physiological data
Xu et al. [38]	Genes	Chaos-based scheme	Protects sensitive genetic data	Narrow biomedical scope
Shi et al. [39]	Medical images	Chaotic image encryption	Strong protection performance	Not EEG-specific
Wang et al. [40]	Bio-electrical signals	Chaotic signal encryption	Improved data security	Lack of EEG adaptation
Murillo et al. [41]	Bio-signals	Ushio chaotic map	Enhanced randomness and security	Requires domain-specific tuning
Lin et al. [42]	EEG	2D chaos-based visual encryption	Lightweight, visual protection	Low-dimensional map, limited key space
Liu et al. [18]	EEG	Hyperchaotic system + authentication	High complexity, strong resistance	Ignores EEG temporal dynamics, lacks adaptivity

3.2. Definition of dual composite IT-ICMIC

The whole structure of ICMIC is straightforward to implement. However, its low sensitivity to initial values is limited, making the output easy to predict. To address these limitations, a dual composite IT-ICMIC is proposed, which introduces inverse trigonometric functions to enhance the nonlinearity and complexity of the chaotic behavior. These functions are chosen for their bounded and nonlinear properties, which improve sensitivity to initial conditions compared to traditional maps. The whole structure is defined as Eq. (2):

$$x_{i+1} = \text{mod}(|\sin(r_2 \times \pi \times \arctan(\sin(\frac{r_1}{x_i}))) + \sin(\frac{r_1}{\sin(r_2 \times \pi \times \arctan(x_i))})|, 1) \quad (2)$$

Where $\text{mod}(\cdot)$ is the modulo operation, r_1 and r_2 are the control parameters in the $(0, +\infty)$ range.

Like other chaotic systems, the map may degrade to periodic or stable behavior if parameters r_1 and r_2 fall outside suitable ranges. In this work, parameters are carefully chosen within intervals where the largest Lyapunov exponent remains positive, ensuring reliable chaotic dynamics.

3.3. Chaotic performance analysis

3.3.1. Bifurcation diagrams

The bifurcation diagram allows for a visual assessment of the behavior exhibited by chaotic maps [46]. Fig. 1 presents the bifurcation diagram of the dual composite IT-ICMIC mapping alongside comparisons with the ICMIC and inverse trigonometric functions. Fig. 1(a) and 1(b) show the bifurcation diagrams of the ICMIC and inverse trigonometric functions, respectively. Since the proposed chaotic maps contain two control parameters, the bifurcation diagrams under different r_1 and r_2 are compared. In Fig. 1(c), the dual composite IT-ICMIC shows a clear white window near 4.5 in the position of r_1 , indicating that it entered a periodic state. In contrast, Fig. 1(d) shows the bifurcation diagram of r_2 as it varies from 2 to 20. The Dual composite IT-ICMIC within this range exhibits almost complete chaos without a clear periodic window, indicating more substantial chaos and instability.

Also, the results suggest that the dual composite IT-ICMIC mapping exhibits significantly stronger chaotic characteristics than the ICMIC and inverse trigonometric functions.

3.3.2. Maximum Lyapunov exponent

The maximum Lyapunov exponent is used to measure the sensitivity of a system to initial conditions and is an essential tool for verifying chaotic maps [47]. Fig. 2 shows the maximum Lyapunov exponent varies with the control parameters r_1 and r_2 . The dual composite IT-ICMIC exhibits a gradually stable chaotic growth trend under the adjustment of two different parameters. Notably, it shows increased instability in the range of higher parameter values, reflecting its enhanced chaotic behavior. Also, Fig. 3 compares the variation trends of the maximum Lyapunov exponent of three chaotic maps at different r values. In contrast, ICMIC and inverse trigonometric methods have lower Lyapunov exponents than dual composite IT-ICMIC and exhibit significant periodic fluctuations at different r values. This result indicates that dual composite IT-ICMIC may have better robustness and encryption performance.

3.3.3. Sample entropy

Sample entropy determines chaos by evaluating the complexity and similarity of data points [48]. Higher sample entropy values typically indicate more complexity and difficulty in predicting. Fig. 4 shows the sample entropy varies with the control parameters r_1 and r_2 . It fluctuates significantly within most parameter values, and the entropy values remain high, indicating that the dual composite IT-ICMIC has high complexity and chaos. From Fig. 5, the proposed method maintains a high level in most ranges of r values and significantly outperforms the other two methods in sample entropy metrics, making the sequence generated by dual composite IT-ICMIC more complex and unpredictable.

3.4. Chaos security analysis

Chaos security analysis can verify the fault tolerance of chaotic systems. Therefore, the dual composite IT-ICMIC is compared with ICMIC and inverse trigonometric maps in terms of EEG encryption

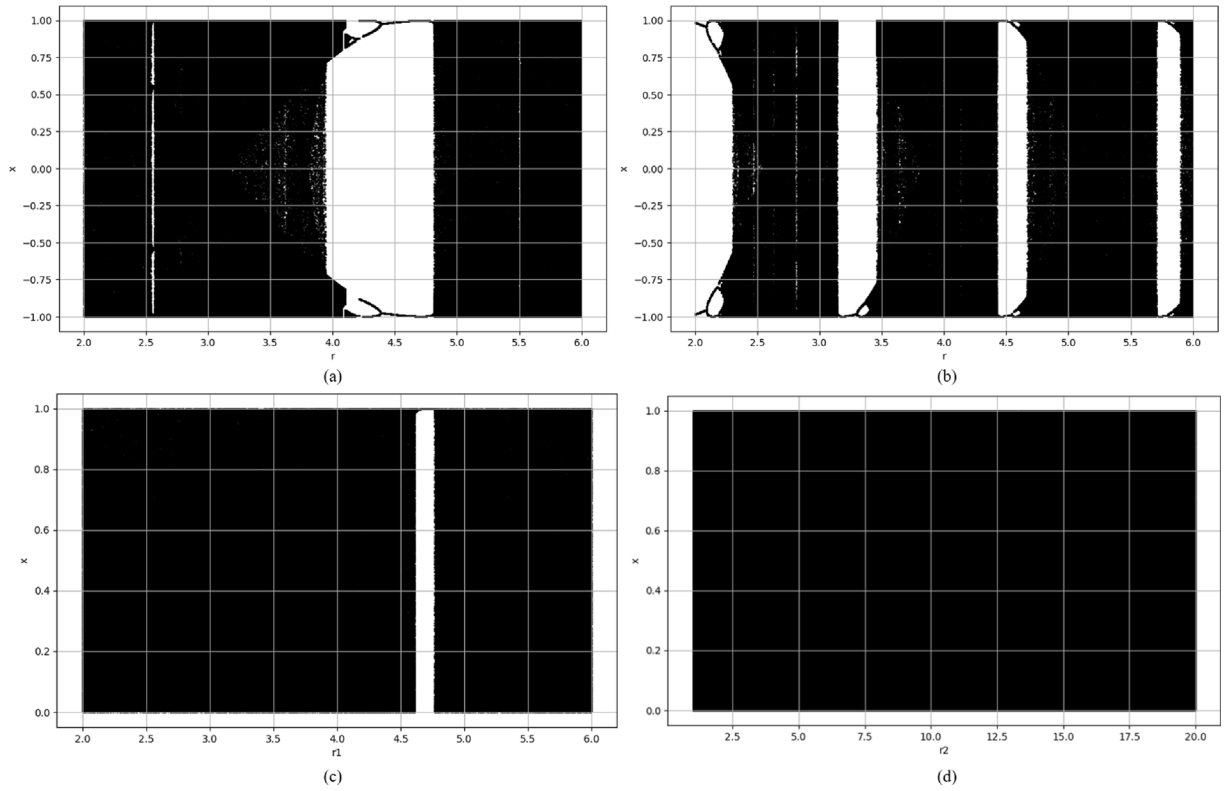


Fig. 1. Bifurcation diagrams. (a) ICMIC, (b) Inverse trigonometric function, (c) Dual composite IT-ICMIC with $r_2 = 18.4723$, (d) Dual composite IT-ICMIC with $r_1 = 5.55$.

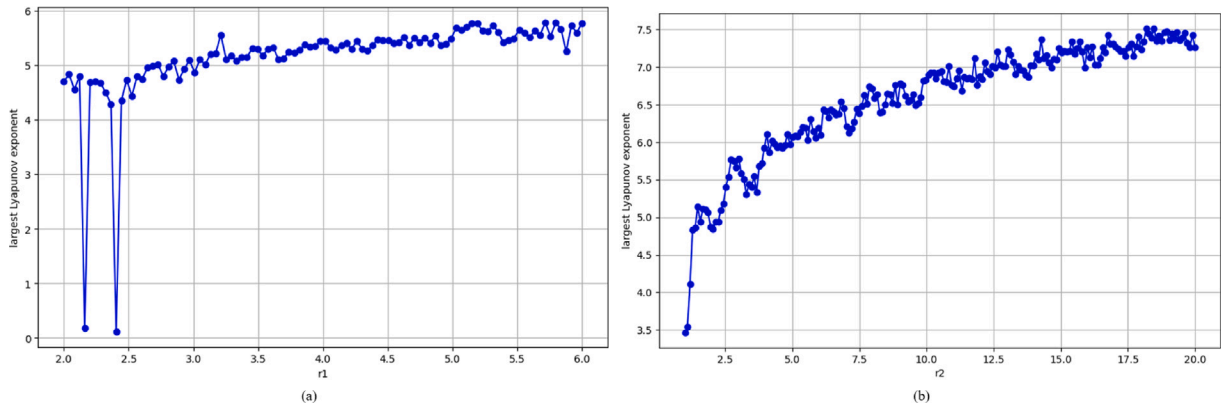


Fig. 2. Lyapunov Exponent. (a) Dual composite IT-ICMIC with $r_2 = 3$, (b) Dual composite IT-ICMIC with $r_1 = 5.7171$.

security. Firstly, the chaotic series is generated through these chaotic maps. Secondly, to illustrate the influence of the chaotic maps, the common encryption method exclusive OR (XOR) is applied to encrypt the EEG signals, which is only used as a baseline to test the performance of the chaotic maps. The tested signal is based on the CHB_MIT EEG Dataset [49] and the initial number is set to 0.7135. Fig. 6 shows the original signal and the encrypted signals by these chaotic maps, which shows that the proposed dual composite IT-ICMIC has better performance.

3.4.1. Correlation analysis

Correlation analysis aims to evaluate the degree of disturbance of signal characteristics by different chaotic mappings, ensuring that the encrypted signal has sufficiently low correlation within adjacent time. Fig. 7 shows the correlation results of EEG signal in adjacent time and Table 3 states the correlation coefficients. It can be seen that the EEG

signals encrypted by the proposed dual composite IT-ICMIC have low temporal correlation adjacent time, while the signals encrypted by the other two chaotic maps maintain relatively high correlation.

3.4.2. Histogram analysis

Histogram analysis is commonly used to analyze the encryption effect and data distribution characteristics of chaotic maps. Fig. 8 shows the histogram results of the encrypted EEG signal based on different chaotic maps. Compared with other chaotic maps, the proposed dual composite IT-ICMIC can make the distribution of values more even.

4. Encryption and decryption of EEG

4.1. Overview

With the increasing demand for personal privacy protection and data security, the EEG signal encryption algorithm based on chaotic

Table 3
Correlation coefficients of XOR encryption based on different chaotic maps.

Chaotic map	Original Signal	ICMIC	Inverse trigonometric	The proposed method
Correlation coefficient	0.8260	-0.0459	0.0488	0.0030

Table 4
Model summary of the proposed EEG encryption framework.

Stage	Operation	Purpose
Input	EEG signal E	Raw physiological data
Pre-processing	Clean E (Filter and normalize)	Artifact suppression and standardized input for encryption
Key generation	Dual composite IT-ICMIC map	Produce chaotic sequences
Adaptive partition	Non-uniform partition	Capture intrinsic EEG dynamics
Permutation	Index sequence M	Reorder EEG samples (scrambling)
Diffusion	Double-step diffusion D_1, D_2	Perturb amplitudes for obfuscation
Output	Encrypted EEG signal	Secured data for transmission

Table 5
Key parameters used in the adaptive non-uniform partitioning process.

Parameter	Symbol	Description
Precision	ϵ	Maximum allowed mean squared error (MSE) for each segment before further partitioning.
Partition number	Y	Record the specific number of signal partitions.
Reconstruction error	err	how well the fitted signal approximates the original EEG signal.
Number of channels	N	N denotes the number of channels for collecting EEG signals.

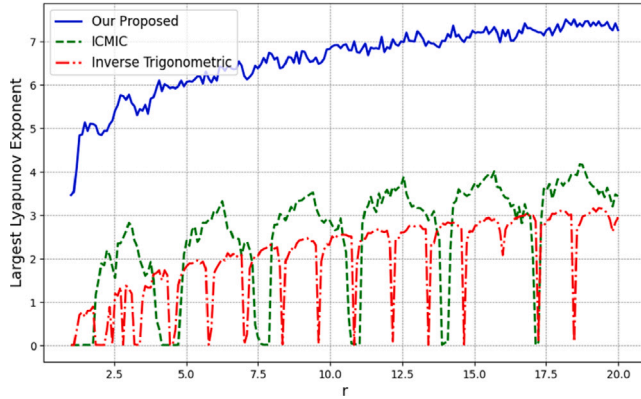


Fig. 3. Comparison of largest LE across three chaotic maps with different r Values.

mapping has gradually received widespread attention. Due to its unique nonlinear characteristics and highly complex dynamic behavior, Chaotic mapping exhibits superior resistance to attacks. It has become a critical technical means for securing EEG signal encryption. However, there are still some issues with existing EEG encryption methods.

Firstly, although chaotic mapping offers significant advantages in encryption, traditional chaotic mapping is subject to potential security risks and limitations in processing EEG signals due to their non-stationary, complex, and highly dynamic characteristics, which hinder the achievement of comprehensive signal protection. Secondly, current encryption strategies often need more in-depth consideration of EEG signal characteristics. They are relatively subjective in design, failing to fully utilize the inherent features of EEG signals, thereby weakening encryption performance and applicability.

To address these issues, an improved EEG signal encryption method is proposed. On the one hand, by designing a chaotic mapping method to enhance the nonlinear complexity and security of the system. On the other hand, in the classic “permutation and diffusion” encryption framework, an adaptive non-uniform partitioning strategy based on EEG signals is introduced, fully utilizing EEG signals’ characteristics during the encryption process, improving the overall encryption performance and anti-attack ability. The flowchart is shown in Fig. 9. To summarize, the model first generates chaotic sequences through

dual composite IT-ICMIC mapping. The initialization is personalized using EEG channel and frequency information, together with adaptive non-uniform partitioning. These chaotic sequences are then applied to scrambling (permutation) and double-step diffusion. This ensures both high complexity and strong adaptability to EEG-specific dynamics. A concise summary of the model stages is provided in Table 4.

4.2. EEG encryption process

Here, the EEG signal is denoted with T sampling points as $E \in \mathbb{R}^{1 \times T}$ and its sampling frequency as f . The number of channels for collecting EEG signals denotes N .

Step 1: EEG Signal Process

Here, the pre-processing of the EEG signal converts the data into $(0, 1)$ with $10 \times e^{-15}$ decimal precision, which is the same in DCLE [19]. The pre-processed EEG data is denoted as P_i , and the calculation is in Eq. (3).

$$P_i = \frac{E_i - (E_{\min} - 0.01)}{E_{\max} + 0.01} \quad (3)$$

Here, the $E_{\min}$ and $E_{\max}$ represent the EEG signal’s minimum and maximum value E .

Step 2: Permutation Series Generation

(1) Non-uniform partition for EEG signal

The introduction of the adaptive non-uniform partition strategy [50] for EEG signals aims to determine the optimal number of partitions for EEG signals while meeting the specific precision requirements. To this end, a cubic fitting function (Eq. (4)) is employed, as higher-order polynomials offer greater flexibility in approximating the non-stationary and nonlinear patterns of EEG signals. As illustrated in Fig. 10, the adaptive non-uniform partitioning progressively divides the original EEG signal into smaller segments whenever the fitting error exceeds the preset threshold. Each segment is then locally approximated using cubic fitting, which maintains reconstruction accuracy. To further improve clarity and reproducibility, the key parameters used in the adaptive partitioning process are summarized in Table 5.

Here, P is taken as an example, and its adaptive non-uniform partition process is described as follows.

Firstly, set the precision ϵ , fitting function $F(x)$, partition number Y .

Secondly, divide the P into two smaller sub-regions P^i , ($i \in 0, 1$).

Thirdly, adopt the fitting function $F(x)$ to fit and reconstruct the P^i by the least square method.

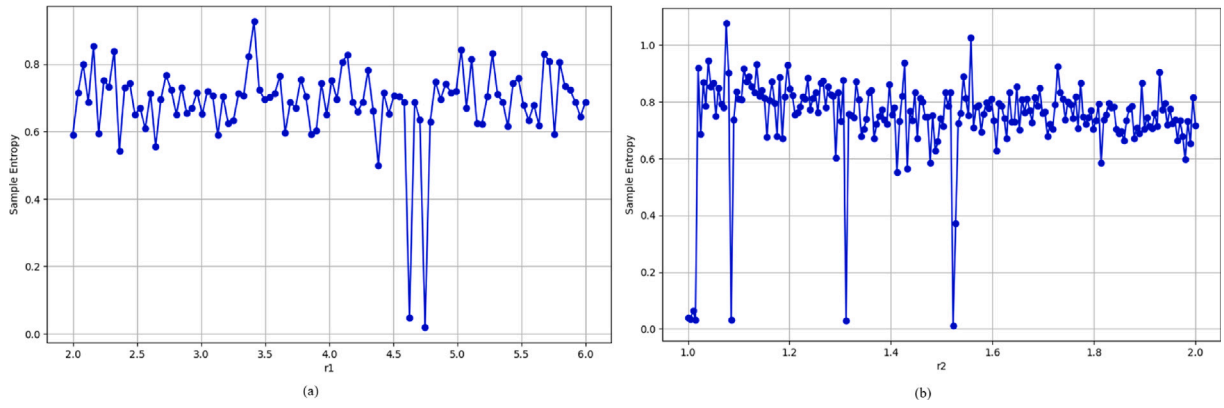


Fig. 4. Sample Entropy. (a) Dual composite IT-ICMIC with $r_2 = 18.4723$, (b) Dual composite IT-ICMIC with $r_1 = 5.7171$.

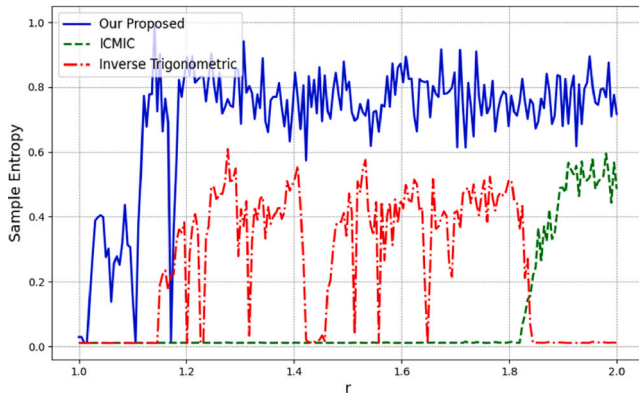


Fig. 5. Comparison of sample entropy across three chaotic maps with different r values.

The goal of the least square method is to minimize the sum of square errors.

$$S = \sum_{m=1}^{\alpha} (P_m^i - (ax_m^3 + bx_m^2 + cx_m + d))^2 \quad (4)$$

where a, b, c , and d are the parameters, x_m denotes the index of the m th sample point in the region P^i , P_m^i represents the actual signal value of the m th sample point with the sub-region P^i .

To minimize the S , take partial derivatives of a, b, c , and d by Eq. (4) in sequence.

$$\begin{cases} \frac{\partial S}{\partial a} = \sum_{m=1}^{\alpha} 2(P_m^i - (ax_m^3 + bx_m^2 + cx_m + d))(-x_m^3) = 0 \\ \frac{\partial S}{\partial b} = \sum_{m=1}^{\alpha} 2(P_m^i - (ax_m^3 + bx_m^2 + cx_m + d))(-x_m^2) = 0 \\ \frac{\partial S}{\partial c} = \sum_{m=1}^{\alpha} 2(P_m^i - (ax_m^3 + bx_m^2 + cx_m + d))(-x_m) = 0 \\ \frac{\partial S}{\partial d} = \sum_{m=1}^{\alpha} 2(P_m^i - (ax_m^3 + bx_m^2 + cx_m + d))(-1) = 0 \end{cases} \quad (5)$$

Eq. (5) can be organized into Eq. (6).

$$\begin{cases} \sum_{m=1}^{\alpha} P_m^i x_m^3 = a \sum_{m=1}^{\alpha} x_m^6 + b \sum_{m=1}^{\alpha} x_m^5 + c \sum_{m=1}^{\alpha} x_m^4 + d \sum_{m=1}^{\alpha} x_m^3 \\ \sum_{m=1}^{\alpha} P_m^i x_m^2 = a \sum_{m=1}^{\alpha} x_m^5 + b \sum_{m=1}^{\alpha} x_m^4 + c \sum_{m=1}^{\alpha} x_m^3 + d \sum_{m=1}^{\alpha} x_m^2 \\ \sum_{m=1}^{\alpha} P_m^i x_m = a \sum_{m=1}^{\alpha} x_m^4 + b \sum_{m=1}^{\alpha} x_m^3 + c \sum_{m=1}^{\alpha} x_m^2 + d \sum_{m=1}^{\alpha} x_m \\ \sum_{m=1}^{\alpha} P_m^i = a \sum_{m=1}^{\alpha} x_m^3 + b \sum_{m=1}^{\alpha} x_m^2 + c \sum_{m=1}^{\alpha} x_m + d n \end{cases} \quad (6)$$

Eq. (6) can be further organized into Eq. (7).

$$\begin{bmatrix} \sum_{m=1}^{\alpha} x_m^6 & \sum_{m=1}^{\alpha} x_m^5 & \sum_{m=1}^{\alpha} x_m^4 & \sum_{m=1}^{\alpha} x_m^3 \\ \sum_{m=1}^{\alpha} x_m^5 & \sum_{m=1}^{\alpha} x_m^4 & \sum_{m=1}^{\alpha} x_m^3 & \sum_{m=1}^{\alpha} x_m^2 \\ \sum_{m=1}^{\alpha} x_m^4 & \sum_{m=1}^{\alpha} x_m^3 & \sum_{m=1}^{\alpha} x_m^2 & \sum_{m=1}^{\alpha} x_m \\ \sum_{m=1}^{\alpha} x_m^3 & \sum_{m=1}^{\alpha} x_m^2 & \sum_{m=1}^{\alpha} x_m & n \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} \sum_{m=1}^{\alpha} P_m^i x_m^3 \\ \sum_{m=1}^{\alpha} P_m^i x_m^2 \\ \sum_{m=1}^{\alpha} P_m^i x_m \\ \sum_{m=1}^{\alpha} P_m^i \end{bmatrix} \quad (7)$$

Eq. (7) can be transformed into Eqs. (8)–(9).

$$A\theta = B \quad (8)$$

$$A = \begin{bmatrix} \sum_{m=1}^{\alpha} x_m^6 & \sum_{m=1}^{\alpha} x_m^5 & \sum_{m=1}^{\alpha} x_m^4 & \sum_{m=1}^{\alpha} x_m^3 \\ \sum_{m=1}^{\alpha} x_m^5 & \sum_{m=1}^{\alpha} x_m^4 & \sum_{m=1}^{\alpha} x_m^3 & \sum_{m=1}^{\alpha} x_m^2 \\ \sum_{m=1}^{\alpha} x_m^4 & \sum_{m=1}^{\alpha} x_m^3 & \sum_{m=1}^{\alpha} x_m^2 & \sum_{m=1}^{\alpha} x_m \\ \sum_{m=1}^{\alpha} x_m^3 & \sum_{m=1}^{\alpha} x_m^2 & \sum_{m=1}^{\alpha} x_m & n \end{bmatrix}, \theta = \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}, B = \begin{bmatrix} \sum_{m=1}^{\alpha} P_m^i x_m^3 \\ \sum_{m=1}^{\alpha} P_m^i x_m^2 \\ \sum_{m=1}^{\alpha} P_m^i x_m \\ \sum_{m=1}^{\alpha} P_m^i \end{bmatrix} \quad (9)$$

According to Eq. (8), the optimal parameters θ' can be obtained by Eq. (10).

$$\theta' = A^{-1}B \quad (10)$$

After obtaining the optimal parameters θ' , the reconstruction loss of the P^i will be calculated. $\theta' = (a, b, c, d)$ here are inherently signal-specific, varying across different recordings, and therefore cannot be reused as universal keys. As a consequence, they are neither stored nor transmitted in plaintext form. Instead, they are computed locally for each dataset instance. Even in the unlikely case that one set of coefficients were intercepted, such information would not generalize to other signals or future sessions. If the reconstruction loss is less than ϵ , Y will increase by 1, but if it is greater than ϵ , the P^i will be further split into two smaller parts (denoted as $P^{i,i'}$, ($i, i' \in 0, 1$)), which $P^{i,i'}$ will be fitted and its reconstruction loss will be compared with ϵ .

Finally, the adaptive non-uniform partition stops when the reconstruction loss of all partition parts is less than ϵ .

The specific operation is listed in Algorithm 1.

Further, the P is summed up to get P_{Sum} . It is used with the obtained number of partitions as the initial value (denoted as W_1) to generate chaotic sequences, which are applied to the permutation step in the encryption process to improve the efficiency and security of the encryption algorithm. The detailed step is shown in Eq. (11).

$$W_1 = \frac{Y + \text{mod}(P_{Sum}, 1)}{3f} \quad (11)$$

Where Y denotes the partition number, P_{Sum} is the accumulated sum of the P , f denotes the sampling frequency.

(2) Generate Permutation Series

After obtaining the initial value W_1 , and given the length T of the EEG signal, a chaotic series is subsequently constructed by dual composite IT-ICMIC (denoted as $C = \{c_i\}_{1 \times T}$). This chaotic series serves as the foundation for generating the permutation sequence (denoted as $M = \{m_i\}_{1 \times T}$). The details are listed according to the Eq. (12).

$$m_i = ([c_i - \lfloor c_i \rfloor] \times 10^{15}) \bmod T \quad (12)$$

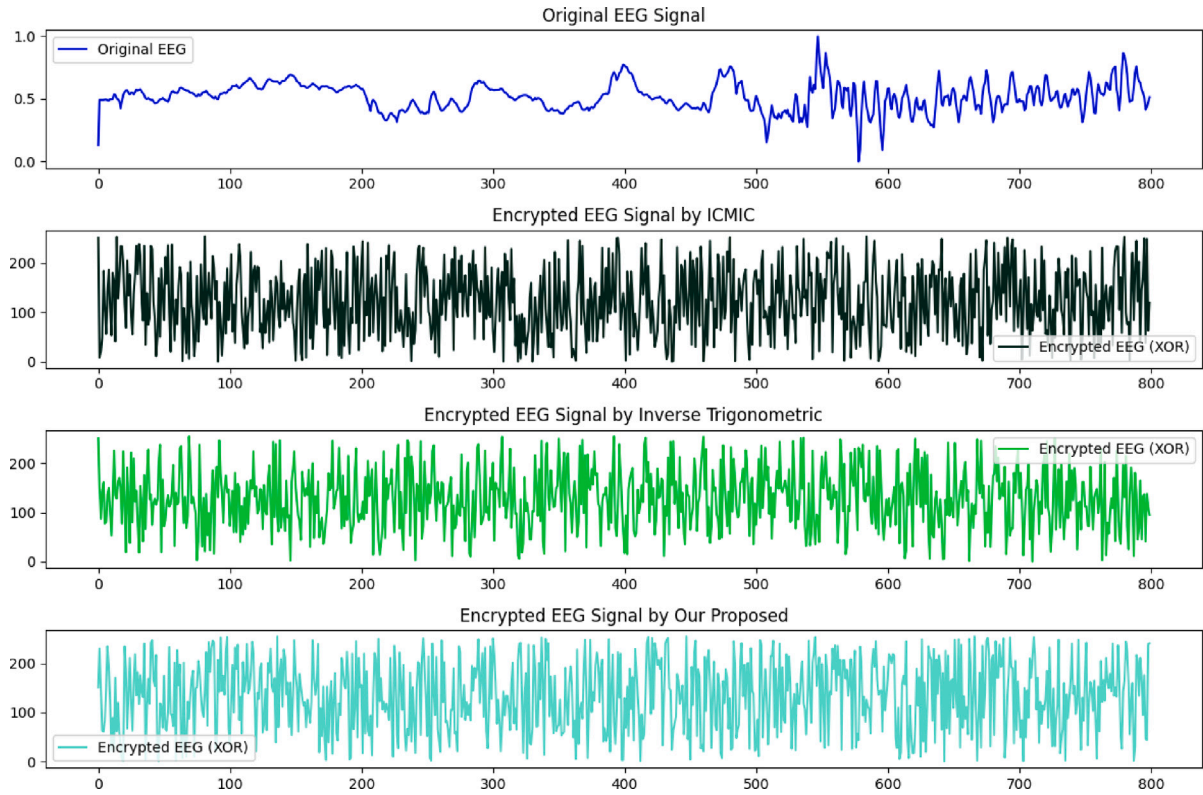


Fig. 6. XOR Encryption of EEG signals based on different chaotic maps.

Algorithm 1 Adaptive non-uniform partition for EEG signal

Require: The pre-processed EEG data P

Ensure: Partition Number Y

```

1: initial  $Y = 0$ 
2: split  $P$  into two smaller sub-regions  $P^i, (i \in 0, 1)$ 
3: compute  $Y$  with  $NU P(P^i)$ 
4: function  $NU P(P^i)$ :
5:   fit  $P^i$  with  $Fit(x) = ax^3 + bx^2 + cx + d$  by least square method
6:   compute  $err^i \leftarrow \frac{1}{a} \sum_{m=1}^a (\hat{P}_m^i - P_m^i)^2$ , where  $\hat{P}_m^i$  is the reconstructed  $P_m^i$ .
7:   if  $err^i \leq \epsilon$  then
8:      $Y = Y + 1$ 
9:   else
10:    split  $P^i$  into two smaller parts  $P^{i,i'}$  ( $i, i' \in 0, 1$ )
11:    compute  $NU P(P^{i,i'})$ 
12:   end if
13: end function

```

where $\lfloor \cdot \rfloor$ denotes the division operation and $\text{mod}(\cdot)$ is the modulo operation, c_i denotes the i th chaotic value generated by the dual composite IT-ICMIC map.

Step 3: Diffusion Series Generation

To enhance the encryption algorithms' effectiveness, EEG signal characteristics are further introduced to generate two diffusion sequences. Firstly, the initial values I_N and I_f are calculated based on the channel number and signal frequency of EEG, respectively. The specific process is listed in Eqs. (13)–(14).

$$I_N = N + \frac{P_{Sum}}{(N + f)} \quad (13)$$

$$I_f = f + \frac{P_{Sum}}{(N + f)} \quad (14)$$

Where N denotes the number of channels. f denotes the sampling frequency.

When obtaining the I_N and I_f , two diffusion sequences generated by dual composite IT-ICMIC (denoted as D_1 and D_2) with the EEG signal length T will be applied sequentially to diffuse the permuted data.

Step 4: EEG Data Permutation

Permutation operations can be used to disrupt the original structure of the EEG signal and increase its unpredictability.

First, M is obtained by Step 2;

Secondly, M is considered as an indexed sequence and used to permute the EEG signal. The detailed process is described in Eq. (15).

$$P'[i] = P[M[i]] \quad (15)$$

where, $i = 1, 2, \dots, T$, P' denotes the permuted EEG signal.

Step 5: EEG Data Diffusion

Diffusion operation can effectively perturb the values of the EEG signal, which helps with data obfuscation and feature hiding. Here, the D_1 and D_2 are adopted to carry out the double-step diffusion on P' . The specific process is shown in Eqs. (16)–(17).

$$D^1[i] = \text{mod}(P'[i] - D_1[i], 1) \quad (16)$$

$$D^2[i] = \text{mod}(D^1[i] + D_2[i], 1) \quad (17)$$

where, $i = 1, 2, \dots, T$, $D^1[i] \in (0, 1)$, $D^2[i] \in (0, 1)$, $D^1[i]$ is the intermediate result after the first diffusion step, $D^2[i]$ is the final result after the second diffusion step.

4.3. EEG decryption process

The decryption process is the reverse step of the encryption process. After receiving the permutation initial value Y , diffusion initial value I_N and I_f , the permutation series M and diffusion series D_1 and D_2

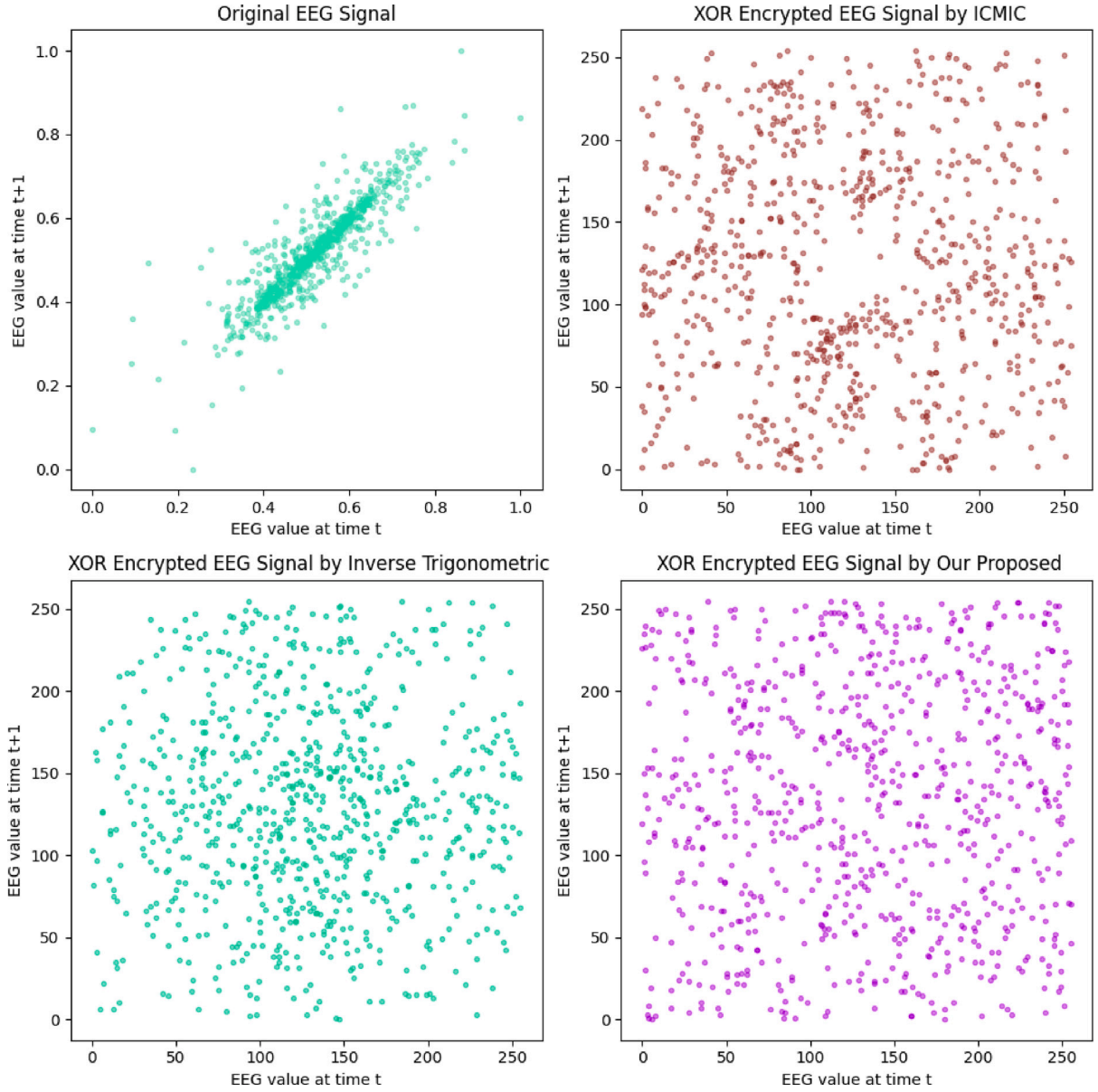


Fig. 7. Correlation of XOR encryption based on different chaotic maps.

can be generated. Then, reverse all steps of the encryption step to get the original EEG data. The detailed step is listed below:

Step 1: Reverse the Decryption Process

Take diffusion series D_1 and D_2 to reverse the diffusion step. The specific process is listed in Eqs. (18)–(19):

$$D^1[i] = \text{mod}(D^2[i] - D_2[i], 1) \quad (18)$$

$$P'[i] = \text{mod}(D^1[i] + D_1[i], 1) \quad (19)$$

where, $i = 1, 2, \dots, T$, $D^1 \in (0, 1)$, $D^2 \in (0, 1)$, $D^1[i]$ is the intermediate result after the first diffusion step, $D^2[i]$ is the final result after the second diffusion step, P' denotes the permuted EEG signal.

Step 2: Reverse the Permutation Process

Take the permutation series M to reverse the permutation step.

The specific process is listed in Eq. (20):

$$P[i] = P'[M[T - i]] \quad (20)$$

Where T denotes the signal length of the EEG signal.

Step 3: Recover the Original EEG Data

Referring to DCLE [19], the pre-processing operations are applied to restore the original EEG data and obtain data E . The process is shown in Eq. (21).

$$E_i = P'_i \times (E_{\max} + 0.01) + (E_{\min} - 0.01) \quad (21)$$

Where the $E_{\min}$ and $E_{\max}$ represent the EEG signal's minimum and maximum value E .

5. Experiments

5.1. Dataset

Four selected EEG datasets are selected for experiments. A detailed summary of these datasets is provided in Table 6.

(1) EEGdenoiseNet [51]

EEGdenoiseNet is a benchmark EEG dataset specifically designed for training and evaluating deep learning-based denoising models. It contains 4514 clean EEG segments with a 512 Hz frequency. Here, 10 EEG signals are selected for the experiment. In this study, 1-second

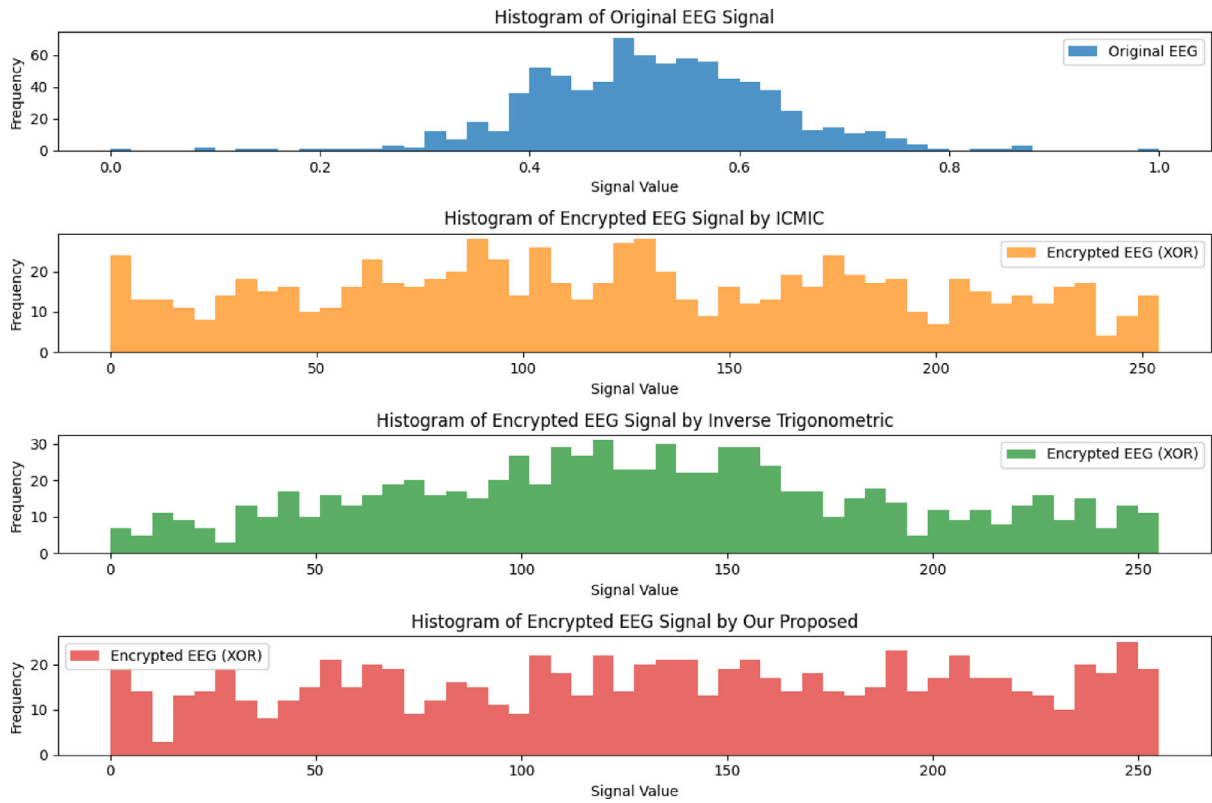


Fig. 8. Histogram of XOR encryption based on different chaotic maps.

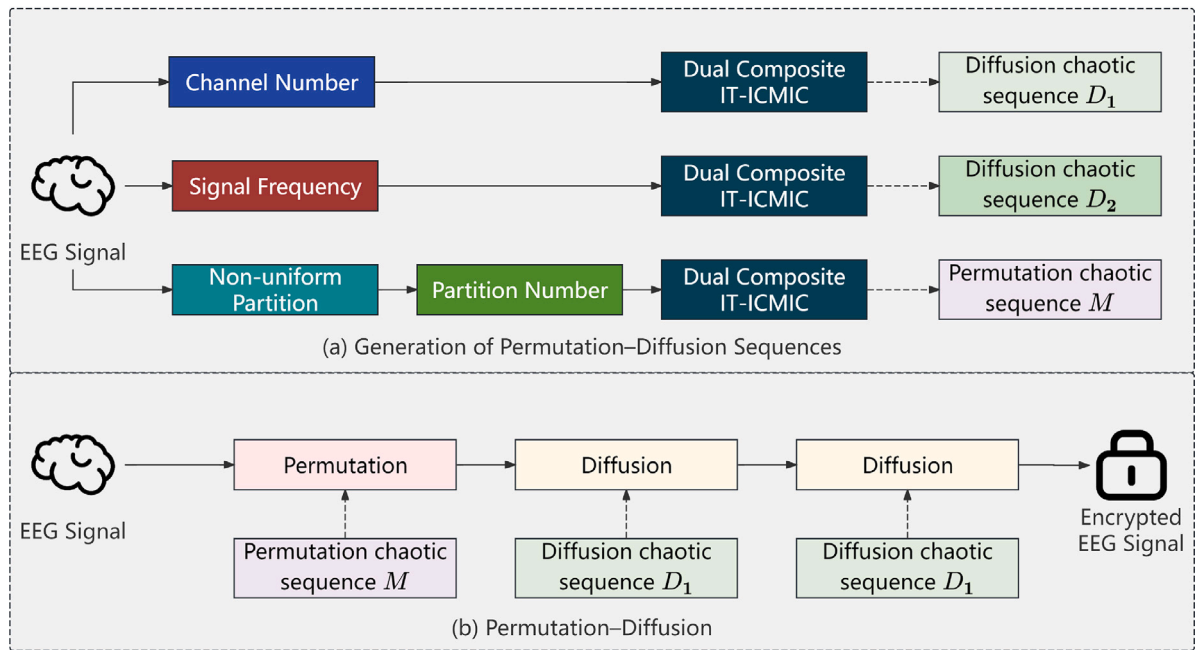


Fig. 9. Flowchart of the proposed EEG encryption method.

segments are adopted, which provide standardized segment-level data for denoising and facilitate rapid evaluation of model performance.

(2) Seed [52]

The SEED dataset is the collection of EEG datasets provided by the Brain-like Computing & Machine Intelligence (BCMI). 62 channels collected the data according to the 10–20 system, and the frequency is down-sampled to 200 Hz. In this experiment, Subject “Seed_20131027” with the channels “PO3”, “POZ”, “PO4”, “PO6”, “PO8”, “CB1”, “O1”,

“Oz”, “O2”, and “CB2” are selected. Specifically, 10 s are selected as the sample duration to provide sufficient temporal resolution for capturing stable emotional states and enable reliable model evaluation.

(3) EEG Motor Imagery Dataset: BCI2000 [53]

Participants completed the different motor imagery tasks while 64-channel EEG was recorded using the BCI2000 system. Here, the S001R01.edf is selected with the first 10 s data, and the frequency is

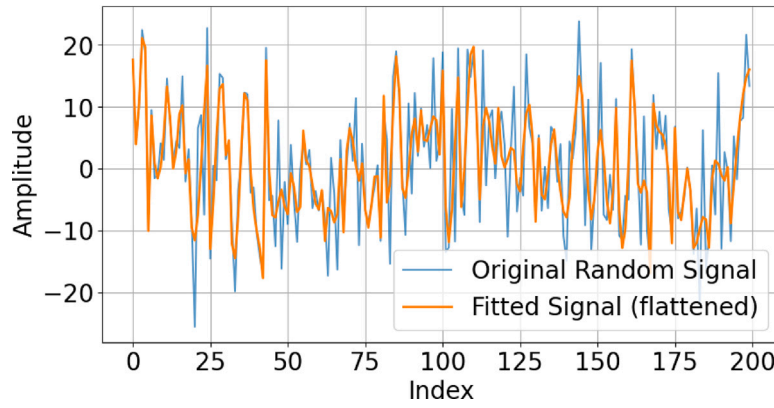


Fig. 10. Example of adaptive non-uniform partition on EEG signal.

Table 6

Summary of datasets with type/task and number of classes.

Dataset	Type/Task	No. of classes
EEGdenoiseNet [51]	Denosing benchmark	– (clean vs. noisy)
SEED [52]	Emotion recognition	3 (positive, neutral, negative)
BCI2000 [53]	Motor imagery	2 (left vs. right hand)
CHB-MIT [49]	Epileptic seizure detection	2 (seizure vs. non-seizure)

160 Hz. The channels are “FC5”, “FC3”, “C5”, “C3”, “P3”, “P5”, “FP1”, “FP2”. It follows the settings of methods to ensure fair comparison.

(4) CHB-MIT EEG Dataset [49]

The CHB-MIT EEG Dataset is a collection of EEG recordings of 22 pediatric subjects. Here, the first 6 s data is selected, and the frequency of the EEG signal is 256 Hz. The channels are “FP1-F7”, “F7-T7”, “T7-P7”, “P7-O1”, “FP1-F3”, “F3-C3”, “C3-P3”, “P3-O1”. The choice of 6 s is consistent with comparative studies, ensuring direct comparability across methods.

5.2. Sensitivity analysis

5.2.1. Dual composite IT-ICMIC chaotic maps

This section will analyze the sensitivity of the dual composite IT-ICMIC chaotic maps from two perspectives: the initial value and the chaotic map parameters.

First, the sensitivity to the initial value W_1 is systematically investigated. Fig. 11(a) shows the original EEG signal, which serves as the reference. The corresponding encrypted signal is presented in Fig. 11(b), confirming that the encrypted signal has no visual similarity to the original signal. Fig. 11(c) displays the decrypted signal without perturbation, where the waveform perfectly overlaps with the original. The mean squared error (MSE) between the original and decrypted signals is as low as 5.58×10^{-33} , which demonstrates nearly lossless reconstruction. To further evaluate robustness, Gaussian noise is injected into W_1 , and the decrypted signal under this perturbation is shown in Fig. 11(d). In this case, visible distortion appears, and the corresponding MSE increases to 0.020. From the results, even very small variations in W_1 lead to drastic divergence from the original, which is also consistent with the PRD-based quantitative analysis. This indicates that the proposed chaotic system is highly sensitive to its initial condition, which helps resist plaintext attacks. Second, the sensitivity is further analyzed for the control parameters r_1 and r_2 . As illustrated in Fig. 12, perturbing either parameter by as little as 10^{-5} leads to a rapid divergence of the generated sequences from the original trajectory. This strong parameter sensitivity further enhances robustness against brute-force and differential attacks.

Table 7

The average of NSCR and UACI results for EEG Dataset.

EEG Dataset	Signal length	NSCR	UACI
EEGdenoiseNet	512	100	33.5619
Seed_20131027	2000	100	33.2113
S001R01	1600	100	33.1893
CHB01	1536	100	33.504

5.3. Anti-differential attack analysis

In this subsection, the resistance of the proposed scheme against differential attacks has been evaluated. A differential attack attempts to exploit small changes in the plaintext and observe their effect on the ciphertext [54]. For EEG encryption, this property is critical, since strong security requires that a single modification in the input leads to widespread and unpredictable changes in the output. The net sample change rate (NSCR) and the unified average change intensity (UACI) are used to measure this effect. NSCR reflects the proportion of changed samples, with an ideal value of 100%. UACI measures the average intensity of the changes, with an ideal value of 33.33%. Values close to these ideals indicate strong resistance to differential cryptanalysis. The equation is defined in Eqs. (22)–(24).

$$U(i) = \begin{cases} 1 & P_1(i) \neq P_2(i) \\ 0 & P_1(i) = P_2(i) \end{cases} \quad (22)$$

$$NSCR = \frac{1}{T} \sum_{i=1}^T U(i) \times 100\% \quad (23)$$

$$UACI = \frac{1}{T} \sum_{i=1}^T |P_1(i) - P_2(i)| \times U(i) \times 100\% \quad (24)$$

Where P_1 and P_2 are the two tested signals.

In the experiment, signals from four EEG datasets (“EEGdenoiseNet”, “Seed_20131027”, “S001R01”, and “CHB01”) were selected. For each signal, one sample was randomly modified 100 times, and the average NSCR and UACI values were calculated. The results in Table 7 show that NSCR remains at 100% for all datasets, and UACI is consistently close to 33%. The consistently high NSCR and near-ideal UACI values confirm that the proposed scheme achieves strong resistance to differential attacks under various EEG datasets.

5.4. Anti-statistical attack analysis

5.4.1. Information entropy

Information entropy [55] is a metric used to measure the degree of uncertainty in a signal, with a theoretical maximum value of 8. For encrypted signals, entropy quantifies their randomness and statistical complexity. A higher entropy value indicates that the encrypted data is

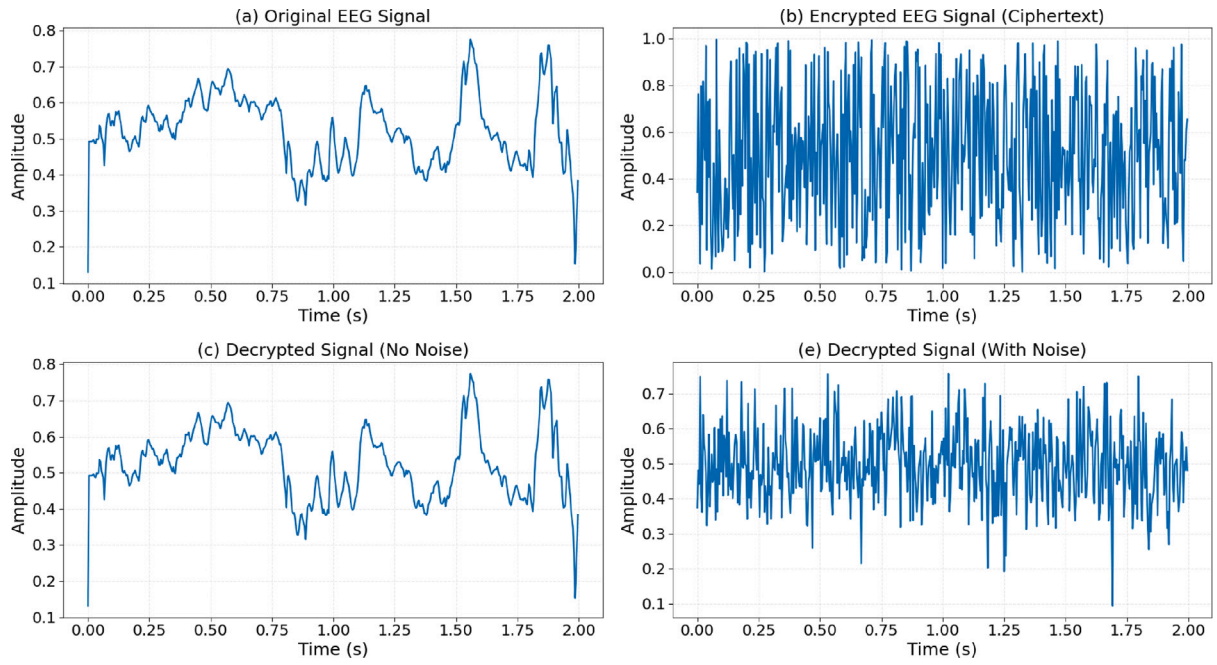


Fig. 11. Sensitivity analysis of initial value W_1 .

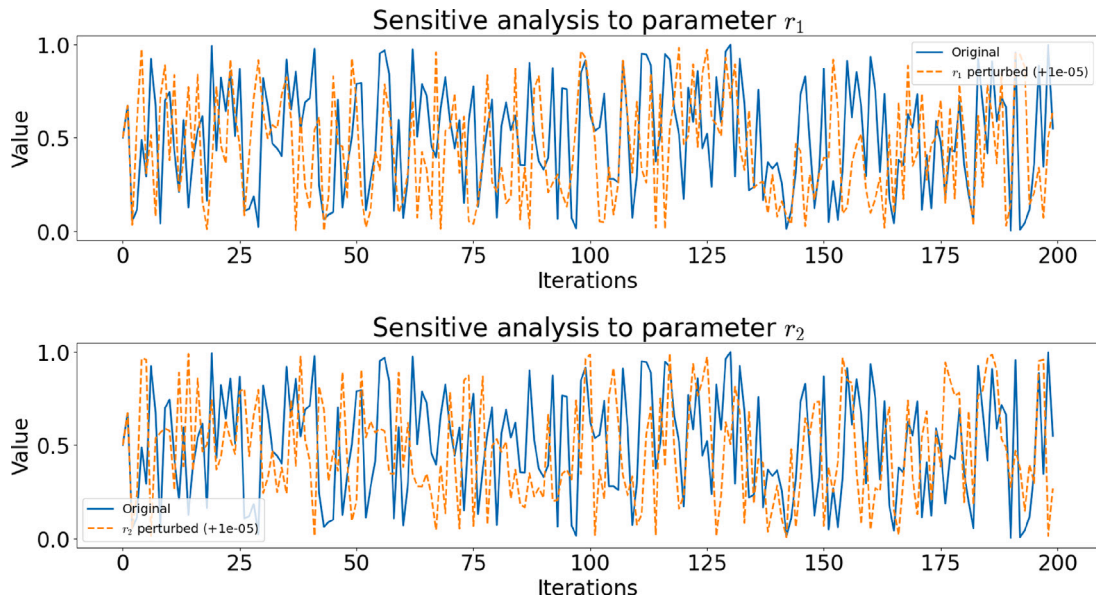


Fig. 12. Sensitivity analysis for control parameters of proposed chaotic maps.

more uniformly distributed and less predictable, which helps to obscure the underlying structure of the original signal. When the entropy approaches its theoretical maximum, it suggests that the encryption algorithm has effectively disrupted statistical patterns. As a result, the ciphertext becomes more resistant to entropy-based statistical attacks, thereby improving the overall security of the encryption method. Its formula is defined in Eq. (25):

$$H(m) = \sum_{i=0}^T p(l_i) \log \frac{1}{p(l_i)} \quad (25)$$

Among them, $p(l_i)$ represents the probability of l_i .

Here, EEG signals are selected from four different datasets “EEG-denoiseNet”, “Seed_20131027”, “S001R01”, and “CHB01” in sequence and encrypt them. Secondly, the range of encrypted data values is adjusted from (0,1) to [0, 255]; finally, the average entropy value is

calculated for the selected EEG signals. The result shown in Table 9 has demonstrated that the encrypted signals have a certain degree of randomness and can resist statistical attack.

5.4.2. Histogram

Histogram was used to analyze the frequency distribution of values in the encrypted EEG signals. A more uniform histogram indicates that the signal values are evenly distributed over the full intensity range. This suggests higher randomness and lower statistical redundancy. In this experiment, such distribution was used as an indicator of encryption effectiveness. A flat histogram reflects a higher degree of complexity, making the signal more resistant to frequency-based statistical attacks [56]. Here, a segment of signal from four datasets “EEGdenoiseNet”, “Seed_20131027”, “S001R01”, and “CHB01” is selected sequentially for histogram analysis. From the results given in

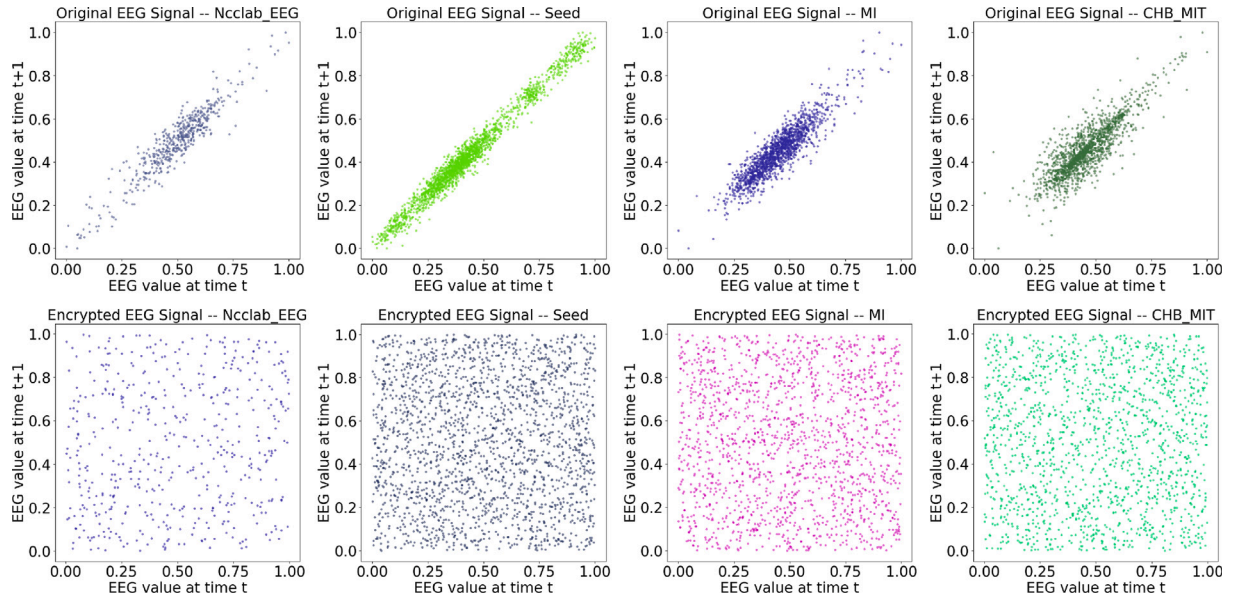


Fig. 13. Correlation of EEG signal encryption based on the proposed method for different datasets.

Table 8

Correlation coefficients of EEG signal encryption based on the proposed method for different datasets.

Dataset	Original signal	Encrypted signal
EEGdenoiseNet	0.9371	0.1164
Seed_20131027	0.9876	-0.0005
S001R01	0.9038	-0.0160
CHB01	0.8473	0.0050

Table 9

Information Entropy results (average values) for EEG Datasets.

EEG dataset	Signal length	Entropy
EEGdenoiseNet	512	7.6093
Seed_20131027	2000	7.8951
S001R01	2400	7.8790
CHB01	1536	7.8487

Fig. 14, The histograms (e)–(h) of the encrypted signals show a uniform distribution, indicating that the signal values are more dispersed after encryption.

5.4.3. Correlation

Signals with high correlation are more likely to expose structural patterns and are vulnerable to statistical attacks. To assess the randomness introduced by the encryption, correlation analysis was applied between the original and encrypted EEG signals. A lower correlation value indicates that the encryption method disrupts the original signal structure more effectively. This suggests higher complexity and better resistance to statistical analysis. In this experiment, correlation was used as a key indicator for evaluating encryption performance. The analysis was conducted on four EEG datasets: “EEGdenoiseNet”, “Seed_20131027”, “S001R01”, and “CHB01”. Fig. 13 shows the correlation results of the original EEG signal and encrypted EEG signal, and Table 8 presents the corresponding correlation coefficients. It can be seen that after encrypting EEG signals using the proposed new encryption methods, the temporal correlation between adjacent time points will decrease.

5.5. Autocorrelation analysis

Autocorrelation analysis aims to analyze the time series by examining the relationship between signal values at different time lags, which

Table 10

Performance Metrics for Different Subjects.

Subject	PRD (Unit: $10 \times e^{-15}$)	R
1	7.796	0.0607
2	38.895	0.0066
3	43.189	0.0473
4	1.823	0.0452
5	31.846	0.0197
6	43.451	0.0040
7	5.454	0.0520
8	38.286	0.0331
9	4.761	0.0568
10	33.010	0.0746

helps in identifying patterns [19]. Firstly, signal segments are selected from four datasets, “EEGdenoiseNet”, “Seed_20131027”, “S001R01”, and “CHB01”, and then encrypted. Secondly, the encrypted EEG signal data are verified via autocorrelation analysis. The results shown in Fig. 15 indicate that the autocorrelation of the original signal is strong, showing apparent periodicity and dependence. While the autocorrelation of the encrypted signal is close to zero, indicating that the encrypted signal has greater randomness.

5.6. Comparative study

To verify the validity of the proposed methods, experiments are conducted on four different datasets: “EEGdenoiseNet”, “Seed_20131027”, “S001R01”, and “CHB01”. The percentage residual deviation (PRD) [57] and Pearson correlation coefficient are adopted as indicators for experimental verification.

The percentage residual deviation (PRD) measures the original and decrypted EEG signal. Lower PRD values indicate higher reconstruction accuracy, with values close to 0% denoting nearly lossless decryption. The calculation is listed in Eq. (26).

$$\text{PRD} = \sqrt{\frac{\sum_{i=1}^T (X_o(i) - X_d(i))^2}{\sum_{i=1}^T X_o(i)^2}} \times 100\% \quad (26)$$

Here, the X_o denotes the original signal and X_d denotes the decrypted signal.

Pearson correlation coefficient (PCC) aims to test the correlation between the original and encrypted EEG signal. The range is $(-1, 1)$,

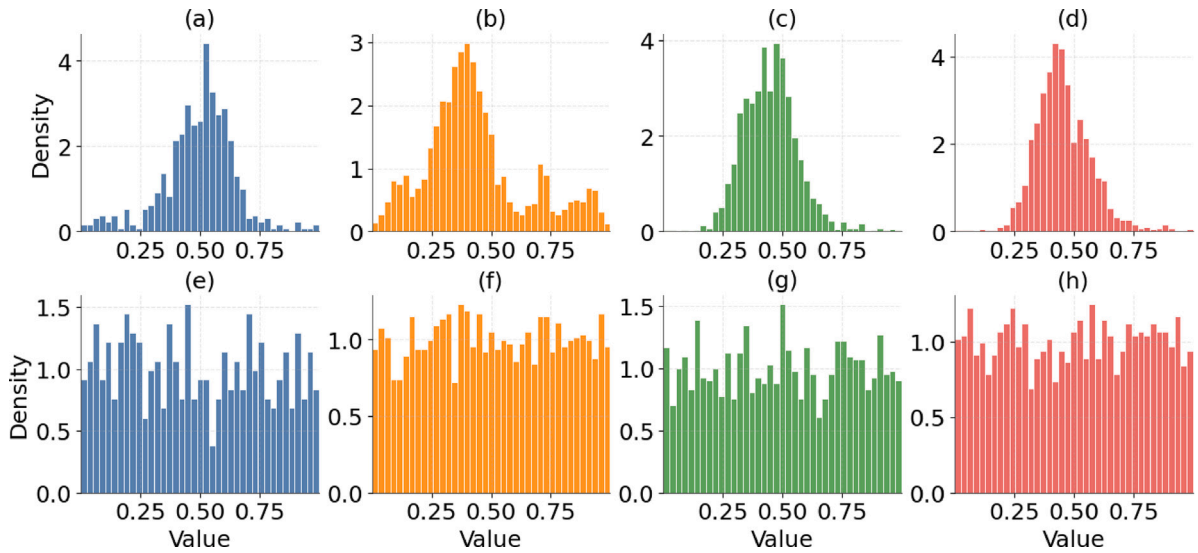


Fig. 14. Histogram of EEG signal before and after encryption. (a) original EEG signal of “EEGdenoiseNet”; (b) original EEG signal of “Seed_20131027”; (c) original EEG signal of “S001R01”; (d) EEG signal of “CHB01” after encryption; (e) EEG signal of “EEGdenoiseNet” after encryption; (f) EEG signal of “Seed_20131027” after encryption; (g) EEG signal of “S001R01” after encryption; (h) EEG signal of “CHB01” after encryption.

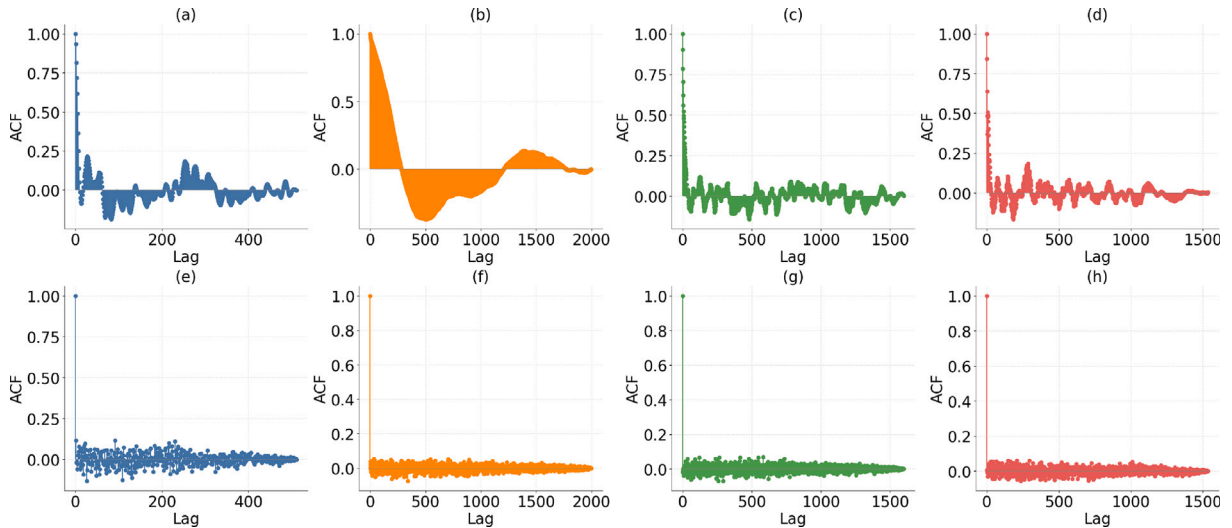


Fig. 15. Autocorrelation of EEG signal before and after encryption. (a) original EEG signal of “EEGdenoiseNet”; (b) original EEG signal of “Seed_20131027”; (c) original EEG signal of “S001R01”; (d) EEG signal of “CHB01” after encryption; (e) EEG signal of “EEGdenoiseNet” after encryption; (f) EEG signal of “Seed_20131027” after encryption; (g) EEG signal of “S001R01” after encryption; (h) EEG signal of “CHB01” after encryption.

and values near 0 imply little correlation. The calculation is described in Eq. (27).

$$R = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^N (x_i - \bar{x})^2 \sum_{i=1}^N (y_i - \bar{y})^2}} \quad (27)$$

Here, the x and y represent the two signals, respectively. $\bar{x}$ and $\bar{y}$ are their corresponding mean values.

Here, the experimental results for “EEGdenoiseNet” and “Seed_20131027” are compared in terms of PRD and PCC. The results are shown in Tables 10–11. Then, the tested results of PRD and PCC for “S001R01” and “CHB01” are depicted in Tables 12–13. Compared to DCLE and K-STBCCS, the proposed method achieves superior results in most channels across both datasets. Specifically, on the S001R01 dataset, the method outperforms in 100% of the channels under the PRD metric and in 62.5% of the channels under the PCC metric. On the CHB01 dataset, the method shows improvements in 75% of the channels under PRD and 75% of the channels under PCC. Further,

we conducted a Wilcoxon signed-rank test across all channels to statistically validate the superiority of the method. For the PRD metric, the proposed approach achieved significantly lower errors compared with DCLE ($p < 0.01$ for both datasets) and K-STBCCS ($p < 0.01$ on S001R01). For the PCC metric, the proposed method significantly outperformed DCLE ($p < 0.05$) and K-STBCCS ($p < 0.01$) on CHB01. Overall, these results demonstrate that the proposed encryption framework not only reduces distortion (low PRD) but also maintains stronger structural similarity (high PCC) compared with baseline methods. These findings clearly highlight the superior performance of the proposed method in secure and reliable EEG signal encryption.

5.7. Robustness verification

Signals are susceptible to noise during transmission. To verify the performance of the proposed method in terms of noise robustness, simulation experiments are conducted based on Gaussian noise with different intensities.

Table 11
Experimental Results for PRD, MSE, and Pearson Correlation Coefficient.

Channel	PRD (Unit: $10 \times e^{-15}$)	R
PO3	1.521	0.0090
POZ	1.805	0.0033
PO4	1.572	0.0357
PO6	6.632	0.0369
PO8	7.086	0.0039
CB1	8.540	0.0274
O1	3.248	0.0054
Oz	9.623	0.0082
O2	7.645	0.0313
CB2	2.955	0.0005

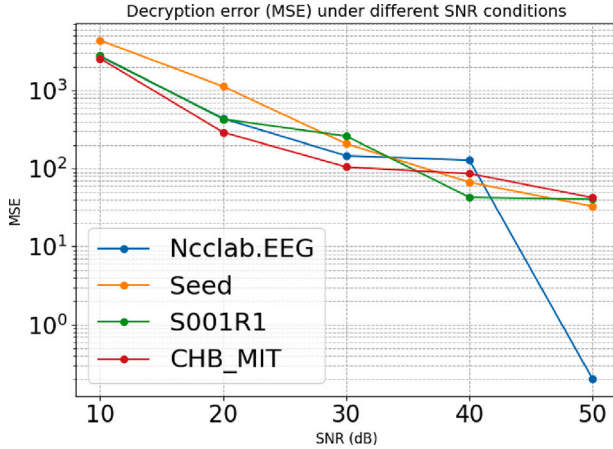


Fig. 16. Decryption error (MSE) of EEG signals under different SNR conditions across four datasets.

Firstly, the EEG signal is encrypted to obtain the ciphertext; secondly, Gaussian noise with varying ratios of signal-to-noise (SNR) is added to the ciphertext, which is then decoded; finally, the MSE between the decoded signal and the original signal is calculated for comparison.

Table 14 shows the MSE between the decrypted EEG signals and the original signals under different Gaussian noise levels. The results have shown that lower SNR leads to higher distortion. Firstly, a 10 dB SNR test was included as a stress-test boundary to probe worst-case behavior. At this level, the noise power rivals or exceeds EEG power, resulting in a sharp increase in MSE. However, in realistic conditions ($\text{SNR} \geq 20$ dB), the MSE decreases quickly and remains low, which confirms the robustness of the proposed method. In addition, Fig. 16 visualizes the results in the form of error curves. The trends clearly show that, across all datasets, the decryption error consistently decreases as the SNR increases. Moreover, the permutation–diffusion transformations ensure reversibility regardless of signal quality. As a result, the encryption–decryption process is inherently independent of the presence of noise or artifacts. This property further supports its applicability to real-world EEG scenarios.

5.8. Resistance to cryptanalytic attacks

The purpose of this experiment is to evaluate the resilience of the proposed encryption framework against advanced cryptanalytic attacks. Specifically, known-plaintext (KPA) and chosen-ciphertext (CCA) attacks are considered to test whether encryption security can be compromised when partial knowledge or oracle access is available to an adversary.

The KPA experiments were conducted on the Seed dataset. For the tested EEG signal, plaintext–ciphertext pairs were generated, and

statistical dependence was evaluated. Without knowledge of the permutation, the correlation between the original and encrypted signal was slight ($\text{PCC} = -0.012$). In the worst-case scenario where the permutation matrix was assumed to be exposed, the similarity between two independent equivalent series remained insignificant ($\text{PCC} = 0.0029$). Prediction attacks using one plaintext–ciphertext pair to reconstruct another also failed, with low correlation ($\text{PCC} = -0.026$). These results confirm that the proposed method is resistant to known-plaintext attacks. For CCA, resistance was tested by introducing controlled perturbations into the ciphertext and analyzing the decrypted outputs. Perturbations with amplitudes ranging from 10^{-12} to 0.5 were applied. Even the smallest changes caused global alterations in the recovered signals, with nearly all positions affected ($\text{NSCR} \approx 100\%$) and average distortion at about one-third of the signal scale ($\text{UACI} \approx 33\%$). The detailed results under different perturbation amplitudes are summarized in Table 15. These findings demonstrate a clear avalanche effect and confirm strong resistance to chosen-ciphertext attacks.

5.9. Time analysis of encryption and decryption

Time delay plays a crucial role in transmitting physiological signals, as it is directly related to the efficiency and performance of information interaction. To test the time performance of the proposed method, EEG data segments are sequentially selected from the EEG dataset, and their overall encryption and decryption times are recorded. The results, shown in Table 16, indicate that the proposed method completes encryption and decryption in a shorter time. It can be observed that the proposed method consistently completes encryption within approximately 0.03–0.05 s and decryption within less than 0.01 s, regardless of variations in signal length or dataset characteristics.

5.10. Qualitative analysis

In addition to the quantitative evaluation, a qualitative case analysis was performed further to demonstrate the effectiveness of the proposed EEG encryption framework. Specifically, one single EEG channel was randomly selected from each dataset (Ncclab_EEG, Seed, MI, and CHB_MIT), and a segment of 500 sampling points was extracted for analysis. These signals were then encrypted using the proposed method. As shown in Fig. 17, the original signals (top row) and their encrypted counterparts (bottom row) were visualized side by side. The encrypted signals exhibit irregularities with no visual similarity to the originals, which highlights the robustness and practical applicability of the proposed framework.

6. Discussion

This study proposes an encryption method for EEG signals based on dual composite IT-ICMIC chaotic maps and an adaptive non-uniform partition algorithm. First, to overcome the complexity limitations of traditional single chaotic mapping methods, the dual composite IT-ICMIC chaotic map approach is introduced, which enhances overall complexity through composite function operations. Second, an adaptive non-uniform partition algorithm is applied to mine the deep dynamic nature within the EEG signal. Furthermore, personalized information is integrated to enhance the encryption's effectiveness. While the proposed method improvements strengthen adaptability and effectiveness, an equally important aspect is the security strength of the method. Low-dimensional chaotic maps often suffer from limited key space and vulnerability to statistical analysis [58,59]. Considering this, the dual composite IT-ICMIC chaotic maps significantly enlarge the key space, which improves sensitivity to initial conditions. Together, these properties offer strong resistance against statistical and differential attacks. Experimental results further demonstrate robustness against practical interference, such as noise contamination and cropping distortions. These findings confirm that the proposed method maintains

Table 12

Comparison of different methods across different channels in “S001R01”.

Channel	FC5	FC3	C5	C3	P3	P5	FP1	FP2
PRD (Unit: $10 \times e^{-15}$)								
DCLE	73.79	70.70	59.56	51.96	58.98	58.98	89.48	88.91
K-STBCCS	57.55	56.31	46.92	38.82	49.30	49.30	67.81	68.74
The proposed method	29.47	49.60	29.47	28.47	38.58	28.72	17.64	62.41
Pearson Correlation Coefficient (PCC)								
DCLE	0.0042	0.0171	0.0053	0.0034	0.0214	0.0112	0.0160	0.0217
K-STBCCS	0.0189	0.0174	0.0134	0.0062	0.0161	0.0158	0.0418	0.0221
The proposed method	0.0029	0.0116	0.0521	0.0010	0.0227	0.0074	0.0070	0.0007

Table 13

Comparison of different methods across different channels in “CHB01”.

Channel	FP1-F7	F7-T7	T7-P7	P7-O1	FP1-F3	F3-C3	C3-P3	P3-O1
PRD (Unit: $10 \times e^{-15}$)								
DCLE	72.25	66.15	76.56	54.96	80.05	84.15	53.38	61.93
K-STBCCS	60.41	51.43	60.64	40.19	63.37	65.96	40.99	49.77
The proposed method	40.46	15.26	29.42	82.37	16.83	92.54	35.01	57.26
Pearson Correlation Coefficient (PCC)								
DCLE	0.0318	0.0233	0.0219	0.0391	0.0260	0.0153	0.0370	0.0331
K-STBCCS	0.0143	0.0662	0.0130	0.0277	0.0451	0.0207	0.0119	0.0071
The proposed method	0.0124	0.0029	0.0012	0.0076	0.0307	0.0222	0.0012	0.0011

Table 14

MSE of decrypted EEG signal and original signal under various signal-to-noise ratios of Gaussian noise.

Dataset	10 dB	20 dB	30 dB	40 dB	50 dB
Ncclab_EEG	2723.559	436.723	145.174	127.735	0.201
Seed	4369.935	1123.326	207.123	66.619	32.702
S001R1	2770.023	429.917	259.276	42.728	40.435
CHB_MIT	2560.760	290.712	104.513	86.054	42.404

Table 15

Results of chosen-ciphertext attack (CCA) evaluation.

Perturbations	NSCR (%)	UACI (%)
10^{-12}	100.0	33.24
10^{-10}	100.0	33.24
10^{-8}	100.0	33.24
10^{-6}	100.0	33.24
10^{-4}	100.0	33.24
10^{-2}	100.0	33.24
10^{-1}	100.0	33.21
0.5	100.0	33.17

Table 16

Encryption time and decryption time of the proposed method for EEG signals.

Dataset	Signal time	Data length	Encrypt time	Decrypt time
Ncclab_EEG	1	512	0.0338	0.0019
Seed	10	2000	0.0428	0.0089
S001R1	10	1600	0.0378	0.0079
CHB_MIT	6	1536	0.0478	0.0069

Table 17

Runtime performance comparison of different methods.

Method	Data	Encryption time (s)	Decryption time (s)
DCLE	EEG (1600)	0.1640	0.1713
EMD	EEG (2000)	0.0749	0.0457
3DMO	ECG (2000)	0.0500	0.0300
The proposed method	EEG (2000)	0.0428	0.0089

both security and reliability. As a result, it is well-suited for EEG data protection in real-world biomedical communication settings. Also, the chaotic sequences are generated for one-time use in each encryption

Table 18

Qualitative comparison of original, encrypted, and decrypted EEG signals across different methods.

Method	Data	MSE	PCC	PRD
DCLE	EEG	8276.23	0.0253	0.04%
EMD	EEG	4412.97	0.0283	6.37×10^{-7}
3DMO	EEG	8456.60	0.0167	0.233%
Liu et al. [18]	EEG	8287.93	0.0377	0.232%
The proposed method	EEG	8694.92	0.0124	33.61×10^{-15}

instance. It will ensure that different EEG sessions are protected with independent sequences. For extended multi-session scenarios, the system can be reinitialized with new seeds, thereby avoiding sequence reuse and avoiding potential risks of statistical or replay-based attacks.

To confirm the practical applicability of the proposed method, a series of comprehensive tests and analyses were conducted to evaluate its effectiveness in various conditions. From the experimental results presented in Tables 12–13, the method proposed demonstrates better performance in both PRD and PCC, indicating improved reconstruction quality after the encryption and decryption of EEG signals. Furthermore, the encrypted data exhibits a lower similarity to the original data, further confirming the method’s superiority in terms of information security. This is likely due to the combined effects of the proposed chaotic mapping and adaptive non-uniform partition algorithm, which, by leveraging the inherent characteristics of EEG signals, not only complicate the encryption process but also enhance its resistance to decryption, ensuring that the encrypted data remains secure.

To explore the time complexity of the proposed method, the theoretical time complexity was first analyzed. The major operations include chaotic sequence generation, adaptive non-uniform partitioning, and permutation–diffusion. Each of these processes scales linearly with the signal length T , and therefore the overall complexity of the framework can be expressed as $O(T)$. It ensures that the method remains computationally efficient even for long EEG recordings. Further, to quantify the computational complexity of the proposed method, the average encryption and decryption time across different EEG datasets are measured (Table 16). The results show that encryption can be completed within approximately 0.03–0.05 s, while decryption requires less than 0.01 s. Further, the runtime performance comparison with other methods is shown in Table 17. The proposed method achieves

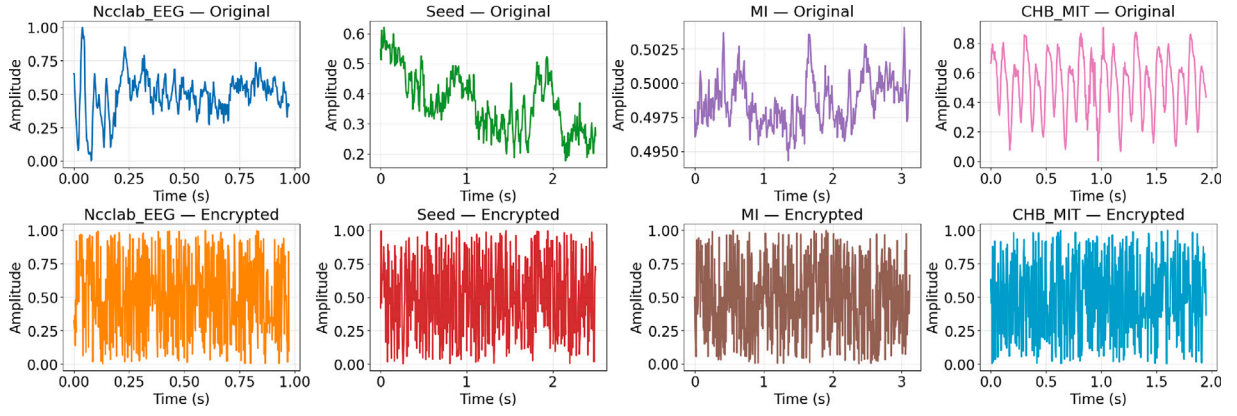


Fig. 17. Qualitative analysis of the proposed EEG encryption method. The top row shows original EEG signals selected from four benchmark datasets (Ncclab_EEG, Seed, MI, and CHB_MIT), while the bottom row presents the corresponding encrypted signals generated by the proposed method.

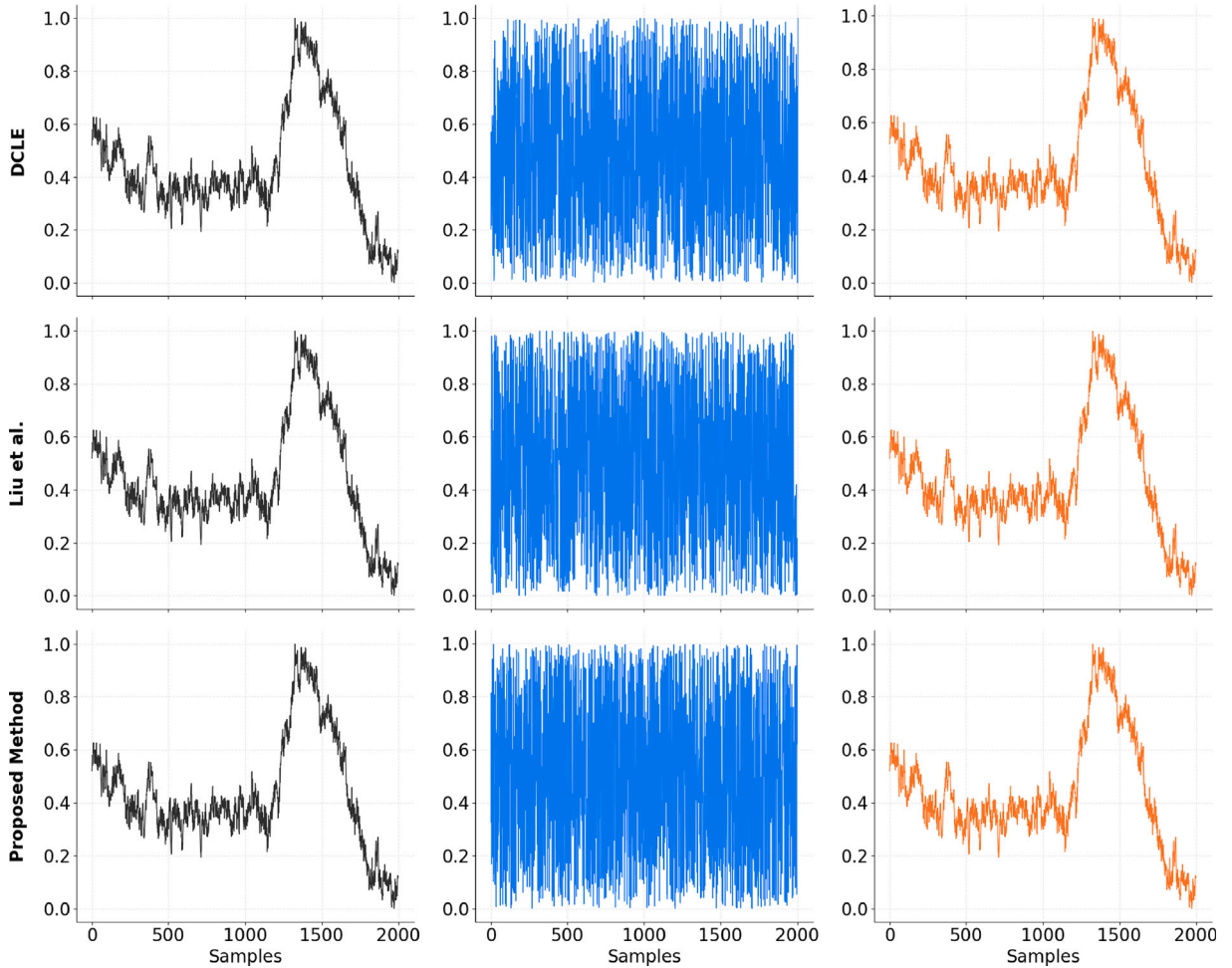


Fig. 18. Qualitative Comparison of Original, Encrypted, and Decrypted EEG Signals Across Different Methods.

competitive encryption time and low decryption time, which is advantageous for real-time biomedical pipelines requiring rapid signal recovery. Both results demonstrate that the proposed method maintains practical runtime efficiency while offering enhanced security. Moreover, the millisecond-level runtime confirms its suitability for online or real-time EEG applications, such as streaming-based monitoring and secure transmission.

To verify the performance of the proposed method, some experimental results of related methods are chosen for comparison. A subset

of EEG signals was selected from the Seed dataset for evaluation. The signals adopt the same pre-processing procedures to maintain consistency. The proposed method and the baseline approaches are then tested under identical experimental conditions. Performance is assessed using three metrics: MSE, PCC, and PRD. MSE and PCC are computed between the original and encrypted signals to evaluate distortion and correlation. PRD is calculated between the original and decrypted signals to measure reconstruction accuracy. Table 18 has demonstrated the comparative results of DCLE [19], EMD [60], 3DMO [61], and Liu

et al. [18]. The results in Table 18 confirm that the proposed method produces stable and reliable performance. Further, to supplement the quantitative results in Table 18, Fig. 18 presents the visualizations of selected comparison methods. The results indicate that the encrypted EEG signals appear largely as noise, with their temporal structure thoroughly masked. Simultaneously, all three methods can decrypt the signals and recover the original waveform morphology. The proposed method yields the encrypted EEG signal with the least visible structure, indicating stronger concealment. Its decrypted signals also align most closely with the original traces. These observations are consistent with the lower PRD values and the higher reconstruction accuracy reported in Table 18.

In general, EEG data analysis relies on feature extraction. However, feature extraction typically produces low-dimensional features. While the feature extraction can reduce computational cost and facilitate subsequent analysis, the reduced input size may also weaken the security strength, particularly in terms of resistance to statistical or brute-force attacks. For this reason, the present study focuses on encrypting raw EEG signals, as their longer sequences enable permutation–diffusion operations to achieve higher complexity and stronger robustness. However, feature-domain encryption remains a potential direction, and future work may explore the hybrid method that combines feature-level encryption with the raw-signal protection method.

However, the presented method still has certain limitations. First, although the new dual composite IT-ICMIC chaotic maps and adaptive non-uniform partition algorithm can improve the encryption effect and security, they also increase the computational complexity. Second, although this encryption method can perform well in encrypting EEG signals, the generalizability of the proposed scheme can be further examined on other biomedical time-series data, such as electrocardiogram (ECG) signals. Future work may involve adapting the method to ECG datasets to validate its robustness under diverse physiological signal characteristics, thereby extending its applicability to healthcare and wearable monitoring scenarios.

7. Conclusion

An EEG signal encryption method based on dual composite IT-ICMIC chaotic maps and an adaptive non-uniform partition algorithm is proposed. The dual composite IT-ICMIC maps enhance sequence complexity and unpredictability, while the adaptive non-uniform partition exploits the intrinsic dynamics of EEG signals to improve adaptability. Experimental results across four EEG datasets confirm the robustness and efficiency of the framework, achieving minimal reconstruction error, ideal NSCR/UACI values, and resilience against noise, cropping, and differential attacks. Furthermore, the scheme ensures real-time performance, with encryption completed within 0.03–0.05 s and decryption in less than 0.01 s. In future work, this framework can be extended to other biomedical signals such as ECG or EMG, as well as real-time brain–computer interface (BCI) applications. Moreover, integrating the proposed encryption scheme with privacy-preserving machine learning and federated learning frameworks may enable secure EEG data sharing and collaborative healthcare applications across institutions.

CRedit authorship contribution statement

Yiran Peng: Writing – original draft, Software, Methodology, Formal analysis, Conceptualization. **Qingqing Hu:** Writing – review & editing, Supervision. **Jing Xu:** Writing – original draft, Supervision. **Yiyao Huang:** Data curation. **Chenheng Deng:** Formal analysis. **U. KinTak:** Project administration.

Funding

No specific funding was received for this research.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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